MARYAM ATHEEQ

Electrical Engineering Student

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WORK EXPERIENCE

System Planning Engineering Co-op Student

Newfoundland Power

Sep – Dec 2026 St. John's, NL

- Prepared net metering applications by reviewing technical specifications and single-line diagrams, performing ***load flow*** and ***short-circuit analyses*** in ***CYME*** to assess grid interconnection feasibility
- Analyzed and updated system load forecasts, infeed forecasts, and infeed load splits using ***MS Excel*** and ***PI DataLink*** to support power system planning activities
- Developed ***Python scripts*** to automate data processing and CYME model updates, streamlining power system modelling workflows and reducing manual effort
- Conducted and updated ***voltage regulation studies*** using ***CYME*** to evaluate feeder voltage profiles and assess distribution system performance
- - Collaborated with the Asset Management team on a ***substation switch risk assessment*** project to support equipment replacement strategies

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Electrical Engineering Co-op Student

Cornwall Electric, a FortisOntario Company

Jan - Apr 2026 Cornwall, ON

- Supported distribution system planning activities for transformers, transclosures, underground cables, and vaults using ***MS Excel*** and ***ArcGIS*** to analyze distribution assets
- Developed utility job drawings in ***AutoCAD*** including pole layouts, clearances and joint-use configurations
- Conducted ***field inspections*** to assess pole conditions and support engineering decisions related to asset maintenance and replacement
- Performed utility pole design and analysis using ***SPIDACalc***, evaluating loading conditions and compliance with utility design standards
- Managed engineering work orders and documentation in ***SAP*** to support construction and planning workflows

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Electrical Engineering Intern

Stantec Inc.

May - Aug 2025 St. John's, NL

- Designed and developed ***electrical power and lighting systems*** creating detailed floor plans, lighting layouts, and photometric renderings in compliance with the Canadian Electrical Code (CEC)
- Performed lighting photometric calculations using ***AGi32*** to support lighting analysis, and conducted short-circuit and arc flash studies using ***ETAP*** with on-site data collection
- Created single-line diagrams using ***AutoCAD*** and developed ***Revit 3D*** models to support building system layouts

EDUCATION

Bachelor of Electrical Engineering, Co-op Program

Memorial University of Newfoundland

Sep 2022 – Present St. John's, NL

Enrolled in Fifth Year (Academic Term 8), Class of 2027

AWARDS & SCHOLARSHIPS

- Guinness World Record, Largest Mosaic
- H. John McDonald Memorial Bursary
- Verafin Inc. Engineering Scholarship
- PEGNL Bursary
- International Entrance Scholarship
- Engineering High Achiever Scholarship

PERSONAL PROJECTS

Security Light System

Designed an **Arduino-based** security light system with motion and light sensors, providing automated and energy-efficient lighting solutions

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Prosthetic Hand

Designed and prototyped a 3D-printed prosthetic hand by integrating micro-servo motors and programming a **Raspberry Pi** to control finger motion

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Portfolio Website

Designed and coded a professional portfolio website using **HTML, CSS** and **JavaScript**

TECHNICAL SKILLS

Revit | AutoCAD | CYME | SAP | ArcGIS | SPIDAcalc | AGi32 | Power Distribution | Lab Equipment | MATLAB | Simulink | LTspice | Circuit Design | IoT Design | Arduino UNO | Project Management | MS Office | 3D Printing | SolidWorks | Onshape | Python | JavaScript